Replicating ATE and ATET Estimates from High-Dimensional Metrics in R

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1 Introduction

This paper replicates the average treatment effect (ATE) and average treatment effect on the treated (ATET) estimates from the 401(k) plan participation application in Section 6.3 of Chernozhukov et al. (2016). The original application illustrates how the `hdm` package implements high-dimensional methods for treatment effect estimation when many controls may be useful for adjustment.

2 Data

The replication uses the `pension` dataset distributed with the `hdm` R package. The outcome is total wealth, and the treatment is 401(k) participation. The control set contains income indicators, age indicators, family size, education indicators, marital status, two-earner household status, defined benefit pension status, IRA participation, and home ownership.

Table 1: Summary Statistics for the 401(k) Analysis Sample

Statistic	Value
Observations	9,915
Covariates	19
Mean total wealth	63,816.85
SD total wealth	111,529.65

3 Methods

The estimating equation follows the specification used by Chernozhukov et al. (2016). I estimate treatment effects with the `rlassoATE` and `rlassoATET` functions from `hdm`. The implementation builds the formula

$$\texttt{tw} \sim \texttt{p401} + \texttt{controls} \mid \texttt{controls},$$

where the controls are selected from the 19 covariates described in the data dictionary. This approach follows the double-selection logic developed in Belloni et al. (2014).

4 Replication Results

Table 2 reports the replicated ATE and ATET estimates generated by the analysis script. The estimates match the target values reported in the original paper up to rounding.

Estimand	Estimate	Std. Error
ATE	10180.09	1930.681
ATET	12628.46	2944.434

Table 2: Replication of ATE and ATET Estimates

5 Challenges in Replication

The main replication challenge was identifying the exact sample, treatment definition, and control set used in the published application. The `hdm` package data and the paper's formula make the computation straightforward once these choices are fixed. A reproducible pipeline is also important because manually copying numeric results into the paper would make it harder to detect changes in data, package behavior, or preprocessing.

6 Takeaways

The replication confirms that the published ATE and ATET estimates can be reproduced from the package data using the stated high-dimensional treatment effect routines. The exercise also shows the value of separating raw input data, preprocessing, analysis, output tables, and the final paper.

7 Extensions

A natural extension is to use 401(k) eligibility as an instrument and replicate the LATE and LATET estimates from the same application. Another extension is to assess sensitivity to alternative control sets or to changes in package versions.

8 References

References

Belloni, A., Chernozhukov, V., and Hansen, C. (2014). Inference on treatment effects after selection among high-dimensional controls. *The Review of Economic Studies*, 81(2):608–650.

Chernozhukov, V., Hansen, C., and Spindler, M. (2016). High-dimensional metrics in R. *The R Journal*, 8(2):185–199.